



Zhuorui YUN

Mechanical Engineering and Robotics
Guangdong Technion-Israel Institute of Technology,
Shantou

+86-15133392728

✉ zhuoruiyun040926@gmail.com

🐙 GitHub Profile

🌐 LinkedIn Profile

EDUCATION

•Guangdong Technion-Israel Institute of Technology

Aug 2023 — Jul 2027

Bachelor of Science in Mechanical Engineering

GPA: 93.5/100 (Top 6%)

- **Relevant Coursework:** Algebra, Linear Systems, Introduction to Robotics, Mechatronics, Introduction to Control, Dynamics

EXPERIENCE

•Bio-inspired Mechatronics Laboratory

Oct 2024 — Present

Undergraduate Researcher

Shantou

- Developing a hybrid aerial-legged robot (quadcopter with a spring-loaded leg), focusing on dynamic modeling, contact-rich simulation, and control
- Designing and evaluating learning-based controllers (PPO/DQN) with control-oriented reward shaping and stability considerations

•Robomaster Kung Fu Team, GTIIT

Jun 2025 — Present

Embedded Control Member

Shantou

- Designed and debugged embedded control pipelines for real-time robot actuation
- Developed asynchronous communication pipelines between onboard computer and STM32 via UART/CAN

RESEARCH PROJECTS

•Quadhopper: Physics-Based Monopod Robot Simulation

Jul 2025 — Present

Physics-based modeling and learning-based control of a quadcopter with a spring-loaded leg

- Built a 2D planar hopper in MuJoCo to validate contact dynamics models and reward formulations; migrated to Isaac Sim 5.1 (Isaac Lab DirectRLEnv) with 256 parallel GPU environments to close the sim-to-real gap
- Applied domain randomization over motor time constants, quadratic thrust curves, mass/inertia, and contact restitution to improve policy robustness
- Modeled hardware-accurate actuator dynamics including first-order lag ($\tau \approx 15$ ms), one-step action delay, and a nonlinear thrust mapping $F(u) = -0.972u^2 + 1.258u - 0.058$
- Trained a unified PPO policy spanning stance and flight phases with an 8-term composite reward; achieved stable in-place repeated jumping with body tilt $< 5^\circ$ and horizontal position error < 0.1 m over 8-second episodes

•Reinforcement Learning Control of Underactuated Pendulum

Aug 2025

Discrete-action RL control for swing-up and stabilization under torque constraints

- Implemented and compared Q-learning, DQN, PPO, and TRPO for pendulum swing-up tasks with discrete action space
- Designed energy-based and constraint-aware reward functions to reduce unnecessary alternating torques
- Visualized learned policies in angle-angular velocity space to evaluate convergence and control smoothness

PUBLICATIONS

- R. Chen, **Z. Yun**, Z. Zhong, Q. Zhang, Y. Or, and M. Liu, “Dynamics-Informed Reinforcement Learning for Agile and Energy-Efficient Locomotion of a Monopodal Hopping Quadcopter,” *Submitted to the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026. **(Under Review)**

TECHNICAL SKILLS AND INTERESTS

Languages: English (Professional Working Proficiency), Chinese (Native)

Programming: Python, C/C++, MATLAB

Frameworks: PyTorch, MuJoCo, Stable Baselines3, NumPy, SciPy

Hardware & Embedded: STM32 Microcontrollers, Oscilloscopes, Sensors (IMU/Encoder), Soldering

Areas of Interest: Robotics Locomotion, Reinforcement Learning, Optimal Control, Quadcopter

POSITIONS OF RESPONSIBILITY

•Undergraduate Peer Tutor

Dept. of Mechanical Engineering, GTIIT

Jun 2024 — Oct 2024

- Led a study group of 10 incoming students during the preparatory semester, providing weekly sessions on study planning and subject-specific guidance
- Facilitated themed experience-sharing meetings each week, covering exam strategies, course selection, and academic adjustment

•Student Representative

Mechanical Engineering Program, GTIIT

Oct 2024 — Present

- Represent the student body in semesterly meetings with Prof. Varenyam Achal (Deputy Dean of Undergraduate Studies); gather and relay student feedback on curriculum design, examination arrangements, and campus life to inform faculty decisions

ACHIEVEMENTS

- Vice Chancellor’s List Top 5% in program2024 – 2025
- The Dean’s List Top 10% of the cohort2023 – 2024
- Most Innovative Project Award Research Poster Competition, GTIITApr 2026
- Chancellor’s Scholarship (Third Class) Awarded for academic excellence2024, 2025